A Critical Data-Driven Analysis of TotalEnergies' ESG & Climate Strategy

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Executive Summary

TotalEnergies SE positions itself as "the major most committed to the energy transition" among international oil companies,[1] claiming Paris Agreement alignment through net-zero commitments, renewable investments, and carbon intensity reductions. This analysis reveals fundamental contradictions between climate ambitions and core business strategy. While leading peers with 22.5% renewable capex and achieving 36% Scope 1+2 emission reductions since 2015, TotalEnergies simultaneously plans 62% oil and gas production growth from 2020 to 2030 ; directly incompatible with IEA Net Zero pathways requiring production declines[2][3]..

The strategy reflects inherent tension: maximizing returns from a $136 billion fossil asset base while claiming climate responsibility. Scope 3 emissions (92% of TotalEnergies' 452 Mt CO_2e footprint) remained flat since 2015 despite production growth[4]. Carbon intensity metrics create accounting illusions: per-unit emissions decline while total emissions plateau.

The climate plan depends on unproven technologies at scale. TotalEnergies targets 10 Mt CO_2 annual CCS capacity by 2030, addressing merely 2% of current emissions. Nature-based offsets (5-10 Mt CO_2 from 2030) face scalability and permanence issues. Low-carbon hydrogen remains commercially nascent. [5]

Legal risks intensify. A French court ruled October 2024 that TotalEnergies committed greenwashing by claiming to be "a major energy transition player" while 97% of revenues derive from fossils.[6] The East African Crude Oil Pipeline (EACOP), projected to emit 34.5 Mt CO_2 annually, faces litigation and international condemnation. [7]

Core Conclusion: TotalEnergies cannot credibly claim Paris alignment while expanding fossil production. The company exhibits relative peer leadership but absolute failure against climate science. With remaining 1.5°C carbon budget at 130 Gt CO_2 (exhausted in ~3 years), an integrated oil major's business model remains incompatible with planetary boundaries. [8] TotalEnergies faces an existential choice: genuine transformation requiring managed hydrocarbon decline, or maintaining profitability while externalizing climate costs to society.

I - Company Overview

Business Model and Scale

TotalEnergies SE operates as a vertically integrated energy major with 2024 revenues of $167 billion, market capitalization of $137 billion, and over 100,000 employees across 120 countries. Operations span exploration and production (upstream), refining and chemicals (downstream), marketing and services, and emerging Integrated Power. [9]

Upstream Oil & Gas: Production of 3.77 Mboe/d in 2024, with 95% of 2030 projected production operational or under development. Major assets include deepwater projects in Brazil, Angola, Nigeria; U.S. shale gas; North Sea fields; and Middle Eastern ventures. [10]

Liquefied Natural Gas (LNG): Global Top 3 producer with 44 Mt annual capacity. Forecasting 50% LNG sales growth by 2030 through Qatar (North Field Expansion), U.S. Gulf Coast (Rio Grande LNG), and Papua New Guinea mega-projects. [11]

Refining & Chemicals: 1.4 million barrels per day capacity across 8 global platforms, generating $25 billion annually in chemical feedstock production.[12]

Integrated Power: Combines renewable electricity generation (26 GW installed), flexible gas plants, battery storage, and EV charging (150,000 charge points). Projected 100-120 TWh electricity production by 2030, becoming cash-flow positive by 2028. [13]

Revenue and Profit Composition

2024 revenue breakdown exposes the green ambition-fossil reality gap:

Segment	Revenue (%)	Operating Income (%)
Exploration & Production	43%	58%
LNG & Gas Trading	40%	28%
Refining & Chemicals	12%	10%
Integrated Power	5%	4% (loss)
Total	100%	100%

Table 1: 2024 Revenue and Operating Income by Segment

This reveals 97% of revenues from fossil operations, renewables contributing 3-5%. Integrated Power, the transition centerpiece, remains unprofitable with positive cash flow delayed until 2028. Management must deliver $40+ billion annual shareholder returns while claiming transition leadership. [14]

Geographic Footprint

TotalEnergies maintains deliberate exposure to geopolitically complex regions offering low-cost reserves: Africa (35% of production), Middle East (25%), Americas (20%), Europe (15%),

Asia-Pacific (5%). This portfolio prioritizes reserve replacement and production growth over decarbonization, targeting 3% annual oil and gas growth through 2030 ; an explicit expansion contradicting phase-down imperatives. [15]

II - Context: Climate Science, Carbon Budgets and IEA Scenarios

The Physical Constraints

The IPCC Sixth Assessment establishes that limiting warming to 1.5°C requires net-zero CO_2 by ~2050 with steep reductions this decade. As of January 2025, remaining carbon budget for 50% probability of 1.5°C stands at ~130 Gt CO_2 (3 years at 40 Gt CO_2 annual emissions) [17]. For 66% probability, the budget shrinks to ~30 Gt CO_2, essentially exhausted in 2025. Even 2°C provides limited breathing room: ~1,200 Gt CO_2 remaining (30 years at current rates).

TotalEnergies' 2024 emissions of 452 Mt CO_2e represent ~1% of global emissions. Planned production increase to 4.7 Mboe/d by 2030 would generate ~500 Mt CO_2e annually including Scope 3. A single oil field or LNG terminal sanctioned today locks in multi-billion tonne emissions over 25-year lifetimes, incompatible with Paris pathways. [19][20]

IEA Net Zero Emissions (NZE) 2050 Scenario

The IEA's NZE represents the authoritative 1.5°C roadmap. [21] Key findings for oil and gas:

1. **No new field development**: "No need for investment in new oil and gas fields" beyond 2021-approved projects. Oil demand peaks 2019, declining 75% by 2050. Gas peaks 2025, falling 60% by 2050. [22]
2. **Accelerated decline rates**: Oil production must fall 4.5% annually; natural decline would produce 8% without capital maintenance. [23]
3. **Scope 1+2 insufficient**: Operational improvements cannot compensate for continued fossil sales. Methane elimination and electrification reduce upstream intensity but not Scope 3. [24]
4. **CCS and hydrogen critical**: NZE requires 5.2 billion tonnes annual CO_2 capture by 2050, necessitating 10 CCS installations monthly from 2030 ; unprecedented pace. [25]
5. **Renewable scale-up**: Solar and wind must grow from ~2,000 GW to ~14,000 GW by 2030 (7x increase), requiring $1.5 trillion annual investment. [26]

IEA Scenario Comparison

Metric	NZE (1.5°C)	APS (1.7°C)	STEPS (2.4°C)
Oil demand 2030 vs 2020	-25%	-12%	+5%
Gas demand 2030 vs 2020	-10%	-6%	+15%
Coal demand 2030 vs 2020	-55%	-30%	-10%
Renewable electricity share 2030	60%	45%	35%
Annual CCS deployment (Mt CO_2)	1,000	500	150

Table 2: IEA Scenario Comparison for 2030 Targets [27][28]

TotalEnergies' Claimed Alignment: The company states strategy aligns with APS (1.7°C) [29], not NZE. This explicitly plans for Paris Agreement failure, targeting 0.2°C above the treaty's 1.5°C limit. Even this claim proves questionable.

Technology Realism

NZE embeds substantial technological optimism. Current global CCS capacity totals ~50 Mt CO_2/year (1% of 2050 requirement). Direct air capture remains prohibitively expensive at $600-1,000/tonne CO_2. BECCS faces land-use competition. NZE assumes these technologies scale 100-fold within 25 years ; unprecedented in energy history. This utopianism provides convenient cover: if future carbon removal offsets present emissions, current expansion becomes defensible. But climate physics offers no credit for optimistic assumptions. [30]

III - TotalEnergies' ESG and Climate Strategy

Net-Zero Commitments: Three-Tier Framework

Tier 1: Operated Facilities (Scope 1+2) ; Net Zero by 2050

Target: Reduce net Scope 1+2 emissions to zero by 2050, with 40% reduction by 2030 vs. 2015. [31]

2024 Progress: Scope 1+2 emissions decreased from 53 Mt CO_2e (2015) to 34 Mt CO_2e (2024) ; 36% reduction. Methane emissions fell 55% vs. 2020, exceeding 50% target early. Upstream intensity reached 17 kg CO_2e/boe, down from 26 kg (-34.6%). [32]

Mechanisms: Energy efficiency ($1 billion investment), flaring elimination (zero routine by 2030), methane leak detection, platform electrification, low-carbon hydrogen in refineries (500 kt/year by 2030), CCS (10 Mt capacity by 2030).[33]

Tier 2: European Operations (Scope 1+2+3) ; Net Zero by 2050

Target: Net-zero across full value chain for European energy products by 2050, with 30% absolute reduction by 2030 vs. 2015. Requires European customers switching from TotalEnergies fossils to electricity, biofuels, or hydrogen. Plan includes 25 GW European renewables by 2030, 100,000 EV charge points, and green hydrogen. [34]

Tier 3: Global Operations (Scope 3) ; Carbon Intensity Reduction

Target: Reduce average lifecycle carbon intensity of global energy products >20% by 2030 vs. 2015, achieving net-zero intensity by 2050. [35]

Critical Distinction: Focuses on carbon intensity (emissions per energy unit), not absolute emissions. TotalEnergies can increase total Scope 3 while meeting intensity targets if production concentrates in lower-carbon products or renewable sales increase. [36]

2024 Status: Lifecycle intensity decreased ~14.7% vs. 2015, tracking toward 17% by 2025. However, absolute Scope 3 emissions remained at 418 Mt CO_2e ; unchanged since 2015. [37]

Capital Allocation: The 75-25 Investment Split

TotalEnergies' capital deployment (annual net capex $16-18 billion): [38]

Investment Category	Annual Capex (2025-2030)	Share
Upstream Oil & Gas	$11.5-12.5B	70-75%
LNG Infrastructure	$4.0-4.8B	23-28%
Refining & Chemicals	$0.5-1.0B	3-6%
Total Fossil	**$16.0-18.3B**	**~75%**
Integrated Power (renewables + storage)	$3.0-4.0B	18-23%
Low-Carbon Molecules (H_2, biofuels)	$0.8-1.2B	5-7%
Total Low-Carbon	**$3.8-5.2B**	**~25%**

Table 3: Capital Allocation 2025-2030

Low-carbon capex grew from $1.8B (2020) to $4.0B (2024), 122% increase. However, fossil spending simultaneously grew from $13.0B to $17.0B (+31%) [39]. October 2024 guidance cut total capex by $1B annually (2026-2030) while maintaining low-carbon at ~$4B. Fossil investments absorbed the entire reduction ; tactical response to $70-80/bbl oil, not strategic reorientation.[40]

Upstream Oil Development: Fossil Expansion

Despite net-zero rhetoric, TotalEnergies aggressively expands production:

GranMorgu (Suriname): FID approved October 2024 for $10.5B offshore development. Production: 220,000 bbl/d from 2028. Lifecycle emissions: ~280 Mt CO_2e over 25 years. [41]

Tilenga and EACOP (Uganda-Tanzania): Six oil fields connected to 1,443-km heated pipeline. Investment: $15B. Projected emissions: 379 Mt CO_2e over 25 years (34.5 Mt annually) ; equivalent to 23 low-income countries combined. [42]

NFE/NFS Qatar LNG: Partnership for world's largest LNG project. TotalEnergies equity in facilities producing 16 Mt LNG annually from 2027, with Scope 3 emissions exceeding 50 Mt CO_2/year. [43]

Ballymore (U.S. Gulf): Deepwater development, 75,000 boe/d, $2B.

Cameia (Angola): Offshore approaching FID, 160,000 boe/d. [44]

Carbon Tracker identifies these as "outside even a 2.5°C outcome." [45] TotalEnergies sanctions ~$25-30B annually in new developments, locking in 20-30 year emission lifetimes incompatible with 2050 net-zero.

Gas and LNG: The "Transition Fuel" Contradiction

TotalEnergies positions gas/LNG as "transition fuels" bridging to renewables. CEO Patrick Pouyanné: "Gas is essential to phase out Asian coal while renewables scale. LNG provides baseload power and reduces emissions 50% vs. coal." [46]

Strategy: Increase LNG sales 50% by 2030 (44 Mt → 66 Mt annually), expand liquefaction through Rio Grande, Qatar, PNG projects. [47]

Reality Check: Lifecycle methane leakage (2-9% of production) eliminates climate advantage over coal when considering methane's 80x warming potency over 20 years. IEA NZE requires gas peak 2025, declining 60% by 2050. TotalEnergies plans opposite: peak 2030, sustained output through 2040. Long-term LNG contracts (15-25 years) face stranded asset risk if gas phase-out accelerates.[48]

Renewables, Batteries, and Hydrogen

Renewable Capacity Trajectory: 5 GW (2020) → 26 GW (2024) → 100 GW target (2030) ; 20% annual growth among fastest globally. Portfolio emphasizes competitive deregulated markets (U.S., Europe, Brazil). [48]

Battery Storage: 4 GWh operational (2024), targeting 20+ GWh by 2030 for intermittency and peak pricing arbitrage. [49]

EV Charging: 150,000 charge points (2024), expanding to 300,000 by 2030. [50]

Low-Carbon Hydrogen: Tender issued for 500 kt/year green or blue hydrogen by 2030 for European refineries, replacing grey hydrogen generating 3 Mt CO_2 annually. Cost premium: ~$3-5/kg vs. $1.50/kg for grey. [51]

Profitability Challenge: Integrated Power generated negative cash flow 2023-2024, breakeven expected 2028. Requires $4B annual investment but produces <5% corporate

EBITDA. Persistent shareholder pressure questions renewable returns vs. high-margin oil/gas. [52]

Governance and Executive Accountability

CEO Long-Term Incentive Plan: 20% of variable compensation tied to carbon intensity reduction (Scope 1+2+3), 15% to renewable capacity growth, 65% to traditional financial metrics (TSR, cash flow). [53]

Say on Climate: TotalEnergies submitted climate strategy to shareholder vote 2021-2023, receiving 91-93% approval. In 2024, shifted to reporting without formal vote, citing EU CSRD regulations. [54]

Investor Pressure: Follow This coalition filed shareholder resolutions demanding Paris-aligned Scope 3 targets, receiving 30-39% support ; substantial minority opposition. Board opposed, arguing existing strategy balances energy security, affordability, and climate. [55]

IV - Critical Analysis: Contradictions and Constraints

The Core Contradiction

TotalEnergies' fundamental strategic contradiction:

1. Promises net-zero by 2050 aligned with (claimed) 1.7°C
2. Plans 3% annual oil/gas growth through 2030, reaching 4.7 Mboe/d
3. Allocates 75% capex to fossils ($110+ billion 2025-2030)
4. Sanctions mega-projects with 25-year emission lock-ins

These objectives require implausible assumptions:

Assumption 1: Scope 3 Declines Despite Production Growth

To achieve 20% intensity reduction while increasing production 25%:

- 2024: 3.77 Mboe/d × 304 kg CO_2e/boe = 418 Mt CO_2e
- 2030: 4.7 Mboe/d × 243 kg CO_2e/boe = 417 Mt CO_2e

Result: Total emissions remain flat, not declining. TotalEnergies merely optimizes carbon content of expanding production. [56]

Assumption 2: Offsets and CCS Provide Neutralization

Plans: 10 Mt CO_2/year CCS capacity by 2030, 5-10 Mt CO_2/year nature-based offsets from 2030. [57]

Reality: Global voluntary carbon markets transact ~200 Mt CO_2 annually, fraught with phantom credits, non-additionality, impermanence. TotalEnergies would require 5-10% of global offset supply at $20-100/tonne costs.

Current CCS capacity equals 50 Mt CO_2 globally. TotalEnergies' 10 Mt target represents 20% of worldwide capacity, requiring 6-8 major facilities in 5 years. Northern Lights (5 Mt) and Bayou Bend (3 Mt) face uncertain customer demand. Even if achieved, 10 Mt addresses only 2% of TotalEnergies' 452 Mt footprint. [58]

Assumption 3: Demand Absorbs Production Increases

TotalEnergies projects oil/gas demand "robust" through 2040 before gradual decline. This conflicts with Paris scenarios:

- IEA NZE: Oil -75%, gas -60% by 2050
- IEA APS: Oil -55%, gas -45% by 2050
- TotalEnergies Forecast: Oil -30%, gas -15% by 2050 [59]

If climate policy accelerates demand destruction, TotalEnergies faces shortfalls and stranded assets.

ESG vs. Fiduciary Obligations

TotalEnergies returned $21B to shareholders in 2023 (dividends + buybacks), maintaining 40%+ cash flow return policy. Annual cash flow: ~$35B (at $80 oil). After shareholder returns ($14-16B), capex ($16-18B), and debt service ($3-5B), available for transformation: near zero. [60]

Pivoting to renewables requires: doubling low--carbon capex to $8-10B annually, accepting 7-10 year paybacks vs. 3-5 for oil, reducing production (cutting cash flows), sacrificing dividend growth (triggering shareholder revolt). [61]

The Innovator's Dilemma: Oil/gas projects generate 15-25% IRRs with established technologies. Renewables offer 8-12% IRRs in competitive auctions. Rationally, management allocates to highest returns, perpetuating fossil dependence. [62]

Greenwashing: Legal Rulings

French Court Verdict (October 2024): Paris Judicial Court ruled TotalEnergies committed "deceptive commercial practices" by claiming to be "a major energy transition player" while 97% of revenues derive from fossils. Findings:

1. Public communications created impression of comprehensive transition
2. Failed to disclose targets depend on unproven CCS/offsets, exclude full Scope 3
3. Continued oil project sanctioning contradicts claims [63]

Penalties: Remove misleading statements within 30 days, display judgment on website 180 days, pay €8,000 damages to each of three NGOs. [64]

EACOP Litigation: East African Crude Oil Pipeline faces multiple challenges including East African Court of Justice appeal, French Duty of Vigilance Law suit by Ugandan plaintiffs (forced evictions, inadequate compensation), and ICJ Advisory Opinion establishing companies "fueling climate crisis risk breaching international law." . [65] Over 25 major banks and 15 insurers publicly declined financing/coverage. TotalEnergies finally secured $3B from Chinese and regional banks [66]

Quantitative Assessment

Emissions Profile (2024)

Emission Category	Annual Emissions (Mt CO_2e)	Share
Scope 1	33	7.3%
Scope 2	1	0.2%
Scope 1+2 Total	**34**	**7.5%**
Scope 3 Upstream (Cat. 1-8)	46	10.2%
Scope 3 Downstream (Cat. 9-15)	372	82.3%
Scope 3 Total	**418**	**92.5%**
Total Carbon Footprint	**452**	**100%**

Table 4: TotalEnergies 2024 Emissions Profile

Source: Tracenable GHG Emissions Database, TotalEnergies 2024 Sustainability Report[67]

Scope 1+2 reductions (36% since 2015) reflect genuine operational improvements but represent only 8% of footprint. Scope 3 (92% of total) remained flat, indicating intensity improvements offset volume increases.

Major Project Emissions Lock-In

Project	Capex ($B)	Lifetime (yrs)	Total Emissions (Mt CO_2)
EACOP/Tilenga (Uganda)	15.0	25	379
GranMorgu (Suriname)	10.5	25	280
Qatar NFE LNG	8.0	30	600+
Ballymore (Gulf)	2.0	20	85
Cameia (Angola)	3.5	25	150
Total	**39.0**	;	**1,494**

Table 5: Major Projects and Lifetime Emissions

These five projects alone lock in 1.5 Gt CO_2 (1.2% of entire remaining 1.5°C budget). The 25-30 year operational lifetimes extend beyond 2050 net-zero deadlines.

VI - Risk Assessment

Regulatory and Litigation Risk

TotalEnergies faces expanding legal exposure:

1. **French Duty of Vigilance Law**: Corporate liability for subsidiary/supply chain environmental/human rights harms [68]
2. **EU CSRD**: Extensive climate disclosure, Scope 3 reporting, scenario analysis requirements; non-compliance carries penalties [69]
3. **EU Green Claims Directive**: Would prohibit unsubstantiated environmental claims, require independent verification [70]
4. **Climate Damages Attribution**: Fossil producers liable for impacts; cases in Hawaii, California, New York survived dismissal [71]

Regulatory Tightening: EU Carbon Border Adjustment, methane regulations, national production bans (France, Denmark, New Zealand). Analyst estimates suggest aggressive policy could reduce market cap 20-40% through stranded assets, carbon pricing, demand destruction. [72]

Transition Risk: Stranded Assets

Carbon Tracker estimates under NZE, 60-80% of oil/gas reserves become unburnable. TotalEnergies' 11.6 billion boe proved reserves would decline to 3-4 billion developable, requiring $25-40B asset impairments. [73]

Demand Destruction Timelines:

Sector	Peak Demand	Decline Rate
Passenger Transport	2019	-3%/yr (EV displacement)
Aviation	2030	-1%/yr post-2035
Petrochemicals	2035	-0.5%/yr (circular plastics)
Shipping	2025	-2%/yr (LNG, H_2, ammonia)
Power Generation	2020	-8%/yr (renewables + storage)

Table 6: Fossil Fuel Demand Destruction by Sector [74]

TotalEnergies' heavy transport fuel exposure (45% of refined products) and LNG (power generation) creates concentrated transition risk.

Reputational Risk and ESG Downgrades

- **MSCI ESG**: Downgraded A to BBB (2023) citing insufficient Scope 3 ambition [75]
- **Sustainalytics**: Elevated to "High Risk" (35.2 score) [76]
- **CDP**: Score declined A- to B (2024) [77]

Consequences: Fossil-free index exclusions limiting institutional access, 1,500+ institutions ($40T AUM) committed divestment, cost of capital increase 50-100 bps. University recruiting boycotts (Oxford, Cambridge) create talent pipeline risks. [78]

Physical Climate Risks

Asset exposures: Gulf of Mexico/North Sea platforms face increasing hurricane intensity, 60% refining capacity in coastal zones vulnerable to sea level rise, African operations experiencing water stress, Arctic projects facing permafrost thaw. TotalEnergies' 2023 scenario analysis estimated €2-3B cumulative physical risk through 2050 under RCP 4.5, rising to €5-7B under RCP 8.5. [79]

Sovereign and Geopolitical Risk

60% of production occurs in countries with elevated political risk: Myanmar (military coup), Uganda (authoritarian governance), Iraq (corruption), Mozambique (insurgency), Russia (sanctions ; $4.1B write-down on Arctic LNG 2), Venezuela (regime uncertainty). Resource nationalism: Uganda revised EACOP terms mid-project demanding higher royalties.

VII - Scenario Analysis (2030 and 2050)

Three scenarios model TotalEnergies' potential trajectories:

Scenario 1: Status Quo Expansion

Assumptions: Current strategy continues (3% annual growth), low-carbon capex remains $4B/year (22-25% total), carbon prices €60-80/tonne Europe, CCS/offsets deployed as planned, 90-100 GW renewables by 2030, global pathway 2.0-2.3°C.

2030 Outcomes:

Metric	Value	vs. 2024
Production (Mboe/d)	4.7	+25%
Renewable Capacity (GW)	35	+289%
Total Emissions (Mt CO_2e)	480	+6%
Carbon Intensity (g CO_2/MJ)	60	-20%
Revenue ($B)	195	+17%

Share Price Index	115	+15%

Table 7: Scenario 1 - Status Quo Expansion (2030)

2050: Production decline post-2035 to 3.0 Mboe/d (2050), Scope 1+2 achieves net-zero through CCS/offsets, Scope 3 declines to 280 Mt CO_2e (missing net-zero), renewables provide 40% revenues, $20-30B stranded asset write-offs (2040-2050).

Climate Impact: Cumulative 2024-2050 emissions ~13 Gt CO_2 (10% of remaining 1.5°C budget). Global temperature 2.1-2.3°C.

Shareholder Returns: TSR 6-8% annually.

Scenario 2: Paris-Aligned Rapid Transition

Assumptions: Aggressive policy (€150-200/tonne carbon Europe, $80-120 globally by 2030), fossil demand collapses (-25% oil, -10% gas by 2030 per NZE), TotalEnergies cancels GranMorgu/Cameia, renewable capex doubles to $8B/year (50% by 2027), global pathway 1.6-1.8°C.

2030 Outcomes:

Metric	Value	vs. 2024
Production (Mboe/d)	3.0	-20%
Renewable Capacity (GW)	55	+511%
Total Emissions (Mt CO_2e)	320	-29%
Carbon Intensity (g CO_2/MJ)	48	-40%
Revenue ($B)	155	-7%
Share Price Index	65	-35%

Table 8: Scenario 2 - Paris-Aligned Rapid Transition (2030)

2050: Production declines to 0.5 Mboe/d (specialty chemicals only), Scope 1+2+3 approaches net-zero (~20 Mt residual), renewables dominate (150 GW, 60% revenues), $50-70B cumulative stranded assets (2025-2040).

Climate Impact: Cumulative emissions ~6 Gt CO_2. Global temperature 1.7°C.

Shareholder Returns: Severe near-term destruction (-35% by 2030), TSR 2-4% annually through 2040, recovering to 5-7% thereafter.

Strategic Imperatives: Immediate production freeze post-2025, triple renewable investment, managed upstream decline, bio-based petrochemicals, capital reallocation from dividends.

Feasibility: Requires shareholder acceptance of 50-60% dividend cuts, activist board representation, potential debt financing. Management faces hostile takeover risk. **Probability: Low (<15%)** absent mandates.

Scenario 3: Delayed Transition and Fossil Supercycle

Assumptions: Climate policy stalls (carbon <€40/tonne, pledges unmet), fossil demand resilience (developing Asia drives growth), TotalEnergies maximizes production (5% annual growth to 2030), low-carbon cut to $2-3B/year, global pathway 2.5-2.8°C.

2030 Outcomes:

Metric	Value	vs. 2024
Production (Mboe/d)	5.5	+46%
Renewable Capacity (GW)	20	+122%
Total Emissions (Mt CO_2e)	570	+26%
Revenue ($B)	220	+32%
Share Price Index	145	+45%

Table 9: Scenario 3 - Delayed Transition / Fossil Supercycle (2030)

2050: Production peaks 2035 (6.0 Mboe/d), declining slowly to 4.5 Mboe/d (2050), Scope 3 remains >400 Mt CO_2e, net-zero claims abandoned for "lower-carbon" positioning, minimal stranded assets, 25 GW renewables.

Climate Impact: Cumulative emissions ~16 Gt CO_2. Global temperature 2.6-2.8°C.

Shareholder Returns: Near-term maximization, TSR 12-15% annually through 2035, followed by volatility.

Moral/Legal Liability: Massive reputational damage, litigation exposure, potential criminal liability for knowingly contributing to catastrophe. **Probability: Moderate (30-40%)** if near-term pressures dominate.

Scenario Comparison Summary

Dimension	Scenario 1	Scenario 2	Scenario 3
2050 Production (Mboe/d)	3.0	0.5	4.5
2050 Renewables (GW)	60	150	25
Cumulative Emissions (Gt CO_2)	13	6	16
Global Warming (°C)	2.1	1.7	2.7
Stranded Assets ($B)	20-30	50-70	5-10

2025-2040 TSR (%)	6-8%	2-4%	12-15%
Paris Alignment	No	Partial	No

Table 10: Scenario Comparison Summary

None achieve genuine Paris alignment while preserving current business model. Scenario 2 approaches compatibility but requires value destruction exceeding management tolerance. Scenario 1 (most probable) represents incremental greening while remaining fossil producer, leading to 2.1°C and managed failure. Scenario 3 demonstrates delayed action's moral hazard: short-term profits at cost of planetary catastrophe.

Final Appraisal

The Fundamental Incompatibility

Thesis: An integrated oil major cannot achieve Paris alignment while remaining a major oil producer. The business model itself ; extracting, refining, selling fossil fuels ; generates emissions requiring elimination.

Mathematical Proof:

1. Paris requires global Scope 3 emissions declining 45% by 2030, approaching zero by 2050
2. Oil/gas combustion generates predictable CO_2: ~0.43 tonnes/barrel oil, ~2.75 tonnes/tonne LNG
3. TotalEnergies' planned 2030 production (4.7 Mboe/d) yields ~420 Mt CO_2e Scope 3 annually
4. Even with 20% intensity reduction, total emissions remain ~380 Mt CO_2e
5. Therefore, production growth contradicts absolute reduction requirements

The strategy relies on three escape routes:

Escape Route 1: Demand Transfer ; If TotalEnergies reduces production, competitors fill gaps. **Rebuttal**: Every actor claims marginal irrelevance, ensuring collective failure. Fiduciary duty cannot excuse climate negligence.

Escape Route 2: Offsetting ; CCS and nature-based solutions neutralize residuals. **Rebuttal**: 10 Mt CCS by 2030 addresses 2% of footprint. Scaling to 400+ Mt proves implausible. Offsets face permanence/integrity crises.

Escape Route 3: Intensity Metrics ; Intensity reductions demonstrate progress. **Rebuttal**: Atmosphere responds to total CO_2, not per-unit efficiency. Intensity improvements within expansion merely optimize destruction rate.

What Genuine Transformation Would Require

Hypothetical Paris-aligned TotalEnergies:

1. **Production Freeze (2025)**: Immediate cessation of new field sanctions
2. **Managed Decline**: Reduce production 5-7% annually, reaching <1 Mboe/d by 2040
3. **Capital Reallocation**: Shift from 75% fossil/25% renewable to 20% fossil/80% renewable by 2030
4. **Dividend Sacrifice**: Cut shareholder returns from $14-16B to $6-8B annually
5. **Renewable Scaling**: Achieve 150+ GW by 2040
6. **Business Pivot**: Transition to renewable electricity, grid storage, green hydrogen, circular petrochemicals, CCS-as-a-service

Financial Implications: Market cap decline 40-60% (2025-2030), margin compression from 25% (upstream) to 8-12% (renewables), TSR near-zero (2025-2035), recovering to 5-7% thereafter.

Barriers: Shareholder revolt, executive self-interest, French government national champion priorities, stranded asset externality (unilateral action transfers value to competitors), technology scale challenges.

Conclusion: Genuine transformation remains theoretically possible but practically implausible absent external forcing (regulation, litigation, systemic crisis). Transition costs exceed stakeholder tolerance; first-mover disadvantage deters unilateral action.

TotalEnergies' Actual Strategic Logic

Alternative Interpretation: TotalEnergies optimizes managed fossil decline while building renewable optionality. The company abandoned Paris alignment, instead positioning for:

1. Extended fossil profitability (2025-2040) maximizing cash from remaining viable demand
2. Regulatory/reputational license through sufficient renewables (20-25% capex)
3. Portfolio hedging with renewables as insurance against faster demand destruction
4. Competitive positioning: outperform Shell/BP in transition metrics, outperform Exxon/Chevron in profitability
5. Geopolitical agnosticism: operate in climate-progressive Europe (renewables) and climate-indifferent Africa/Middle East (fossil expansion)

This strategy accepts 2.0-2.3°C warming as inevitable, abandons 1.5°C as unrealistic, prioritizes corporate survival over planetary boundaries. **From shareholder value perspective, it may be optimal. From climate justice perspective, it is catastrophic.**

The Peer Context

TotalEnergies' relative leadership reflects sector-wide failure:

- BP: Abandoned 40% production reduction target (2020), now increasing fossil investments 20%
- Shell: Scaled back renewables, focusing LNG and "transition" petrochemicals

- Equinor: Maintains ~26% low-carbon capex but continues North Sea expansion
- Exxon/Chevron: Token renewables (<2%), explicit fossil maximization

Being "least bad" among companies actively destabilizing climate offers no legitimacy when carbon budgets near exhaustion.

Final Verdict

Genuine Strengths: 22.5% low-carbon capex (highest among majors except Equinor), 26 GW renewables (real assets), 36% Scope 1+2 reduction, methane elimination leadership (-55%), pioneering CCS, climate governance integration.

Fundamental Contradictions: 3% annual oil/gas growth through 2030, $110B fossil capex (2025-2030) locking in 1.5 Gt CO_2, EACOP (379 Mt CO_2) approved post-2020 commitments, Scope 3 flat since 2015, reliance on unproven CCS/offsets (addresses 2% of footprint), intensity metrics masking absolute stagnation, greenwashing conviction.

Conclusion: TotalEnergies exhibits **performative transition** ; sufficient gestures toward climate responsibility to maintain social license while fundamentally preserving fossil business model. The company cannot credibly become a "responsible energy major" while expanding oil/gas production. The phrase itself embeds contradiction: "energy major" status depends on scale incompatible with "responsible" action under climate constraints.

A More Honest Framing: TotalEnergies could rebrand as "transitioning fossil fuel company attempting to balance climate risks with near-term profitability while acknowledging incompatibility with 1.5°C pathways." This, though commercially toxic, would represent intellectual honesty currently absent.

The Systemic Diagnosis: TotalEnergies' failure reflects capitalism's structural inability to self-regulate existential externalities. Corporate fiduciary duty, quarterly earnings pressures, and competitive dynamics incentivize short-term extraction over long-term planetary stability. Individual corporate virtue cannot overcome systemic misalignment.

Prescription for Change (requiring state intervention):

1. Carbon price floor (€150-250/ton) making fossil production unprofitable
2. Managed supply decline mandates capping production with phased reductions
3. Renewable portfolio standards for energy companies (50% by 2030, 90% by 2040)
4. Climate liability frameworks exposing companies to Scope 3 damages
5. Just transition financing for worker retraining and stranded asset write-offs

Absent regulatory forcing, TotalEnergies will continue optimizing fossil profitability while greenwashing ; a strategy ensuring climate catastrophe while preserving quarterly dividends.

References

[1] https://totalenergies.com/system/files/documents/totalenergies_sustainability-climate-2025-progress-report_2025_en.pdf

[2] https://totalenergies.com/system/files/documents/totalenergies_2024_Results_and_2025_Objectives_presentation.pdf

[3] https://www.enerdata.net/publications/daily-energy-news/totalenergies-targets-3year-increase-oil-and-gas-production-2030.html

[4] https://tracenable.com/company/totalenergies/ghg-emissions

[5] https://totalenergies.com/system/files/documents/2024-03/totalenergies_sustainability-climate-2024-progress-report_2024_en.pdf

[6] https://www.climateinthecourts.com/totalenergies-misled-consumers-court-rules-in-historic-greenwashing-judgment/

[7] https://cri.org/uganda-cnooc-totalenergies-oil-project-fueling-serious-abuses/

[8] https://climatechangetracker.org/climate-change-progress/current-remaining-carbon-budget-and-trajectory-till-exhaustion

[9] https://totalenergies.com/info/esg

[10] https://www.akm.ru/eng/news/totalenergies-has-published-a-strategy-and-forecast-for-2026-2030/

[11] https://www.investing.com/news/company-news/totalenergies-cuts-capex-announces-75-billion-savings-program-93CH-4261078

[12] https://totalenergies.com/features/decarbonization-actions

[13] https://totalenergies.com/news/press-releases/2025-strategy-and-outlook-presentation

[14] https://www.morningstar.com/news/dow-jones/202509296462/totalenergies-to-cut-costs-while-boosting-oil-gas-output-through-2030

[15] https://reclaimfinance.org/site/en/2025/03/11/totalenergies-the-usa-gas-rush/

[16] https://www.ipcc.ch/sr15/chapter/chapter-2/

[17] https://climate.leeds.ac.uk/news/scientists-find-three-years-left-of-remaining-carbon-budget-for-1-5c/

[18] https://climatechangetracker.org/climate-change-progress/current-remaining-carbon-budget-and-trajectory-till-exhaustion

[19] https://www.nature.com/articles/s41558-023-01848-5

[20] https://carbontracker.org/reports/paris-maligned/

[21] https://zerocarbon-analytics.org/policy/what-does-the-iea-net-zero-scenario-say/

[22] https://www.wri.org/insights/5-things-know-about-ieas-roadmap-net-zero-2050

[23] https://mn350.org/2021/11/achieving-net-zero-emissions-by-2050-the-iea-1-5c-scenario/

[24] https://totalenergies.com/sustainability/climate-and-sustainable-energy/reducing-our-emissions

[25] https://zerocarbon-analytics.org/policy/what-does-the-iea-net-zero-scenario-say/

[26] https://www.iea.org/reports/net-zero-by-2050

[27] https://zerocarbon-analytics.org/policy/what-does-the-iea-net-zero-scenario-say/

[28] https://www.cnbc.com/2025/11/13/what-now-for-peak-oil-unpacking-the-ieas-shift-on-fossil-fuel-demand.html

[29] https://totalenergies.com/system/files/documents/2023-03/2023_TotalEnergies_Strategy_Sustainability_Climate_Presentation.pdf

[30] https://zerocarbon-analytics.org/policy/what-does-the-iea-net-zero-scenario-say/

[31] https://totalenergies.com/sustainability/climate-and-sustainable-energy/reducing-our-emissions

[32] https://totalenergies.com/news/press-releases/totalenergies-publishes-its-sustainability-climate-2025-progress-report-and

[33] https://totalenergies.com/features/decarbonization-actions

[34] https://totalenergies.com/system/files/documents/2024-03/totalenergies_sustainability-climate-2024-progress-report_2024_en.pdf

[35] https://totalenergies.com/sites/g/files/nytnzq121/files/documents/2022-01/2021_07_08_MEDEF-French_Busiess_Climate_Pledge_TTE_EN.pdf

[36] https://follow-this.org/totalenergies-bet-on-gas-means-no-big-emissions-cut-by-2030/

[37] https://tracenable.com/company/totalenergies/ghg-emissions

[38] https://totalenergies.com/system/files/documents/totalenergies_2024_Results_and_2025_Objectives_presentation.pdf

[39] https://totalenergies.com/news/press-releases/2025-strategy-and-outlook-presentation

[40] https://www.investing.com/news/company-news/totalenergies-cuts-capex-announces-75-billion-savings-program-93CH-4261078

[41] https://totalenergies.com/news/press-releases/suriname-totalenergies-announces-fid-granmorgu-project

[42] https://cri.org/uganda-cnooc-totalenergies-oil-project-fueling-serious-abuses/

[43] https://www.akm.ru/eng/news/totalenergies-has-published-a-strategy-and-forecast-for-2026-2030/

[44] https://carbontracker.org/oil-and-gas-companies-are-way-off-track-from-paris-agreement-goals-finds-new-combined-alignment-scorecard/

[45] https://carbontracker.org/oil-and-gas-companies-are-way-off-track-from-paris-agreement-goals-finds-new-combined-alignment-scorecard/

[46] https://energydigital.com/articles/sustainable-development-at-heart-of-totalenergies-strategy

[47] https://reclaimfinance.org/site/wp-content/uploads/2023/10/Assessment-of-TotalEnergies-Climate-Plan.pdf

[48] https://totalenergies.com/news/press-releases/2025-strategy-and-outlook-presentation

[49] https://esgpost.com/totalenergies-buys-50-of-ephs-e10-6bn-power-portfolio/

[50] https://totalenergies.com/investors/investors-presentations

[51] https://totalenergies.com/news/news/hydrogen-decarbonization-refineries-europe

[52] https://totalenergies.com/news/press-releases/2025-strategy-and-outlook-presentation

[53] https://totalenergies.com/system/files/documents/totalenergies_universal-registration-document-2024_en.pdf

[54] https://totalenergies.com/investors/shareholders-meetings

[55] https://collaborate.unpri.org/group/18566/home

[56] https://reclaimfinance.org/site/wp-content/uploads/2023/10/Assessment-of-TotalEnergies-Climate-Plan.pdf

[57] https://totalenergies.com/features/decarbonization-actions

[58] https://enkiai.com/totalenergies-carbon-capture-initiatives-for-2025-key-projects-strategies-and-market-impact

[59] https://totalenergies.com/sites/g/files/nytnzq121/files/documents/2023-11/TotalEnergies_Energy_Outlook_2023_EN.pdf

[60] https://totalenergies.com/system/files/documents/totalenergies_2024_Results_and_2025_Objectives_presentation.pdf

[61] https://totalenergies.com/investors/publications-and-regulated-information/regulated-information/annual-financial-reports

[62] https://carbontracker.org/stranded-strategies/

[63] https://www.reuters.com/sustainability/boards-policy-regulation/french-court-rules-totalenergies-misled-consumers-with-carbon-neutrality-claims/

[64] https://www.esgtoday.com/totalenergies-pushes-back-against-greenwashing-accusations-after-court-ruling/

[65] https://drilled.media/news/eacop-lawsuit

[66] https://www.business-humanrights.org/en/latest-news/projet-eacop-de-totalenergies

[67] https://tracenable.com/company/totalenergies/ghg-emissions

[68] https://totalenergies.com/sustainability/climate-and-sustainable-energy/ghg-emissions

[69] https://www.business-humanrights.org/en/big-issues/corporate-legal-accountability/french-duty-of-vigilance-law-faq/

[70] https://finance.ec.europa.eu/capital-markets-union-and-financial-markets/company-reporting-and-auditing/company-reporting/corporate-sustainability-reporting_en

[71] https://environment.ec.europa.eu/topics/circular-economy/green-claims_en

[72] https://carbontracker.org/reports/stranded-assets-danger-zone/

[74] https://carbontracker.org/stranded-strategies/

[75] https://www.iea.org/reports/world-energy-outlook-2023

[76] https://www.msci.com/our-solutions/esg-investing/esg-ratings-climate-search-tool/issuer/totalenergies-se/IID000000002158488

[77]https://www.cdp.net/en/responses?utf8=%E2%9C%93&queries%5Bname%5D=TotalEnergies

[78] https://peopleandplanet.org/fossil-free

[79] https://totalenergies.com/system/files/documents/2023-03/2023_TotalEnergies_Universal_Registration_Document.pdf